Activity 2

Visit a nearby nursery and observe how *Sphagnum* is used in horticulture and make a note on it.

17.4 Pteridophytes

Pteridophytes are the first true land plants with xylem and phloem. Hence, they are called vascular cryptogams. The main plant body is differentiated into true root, stem and leaves.

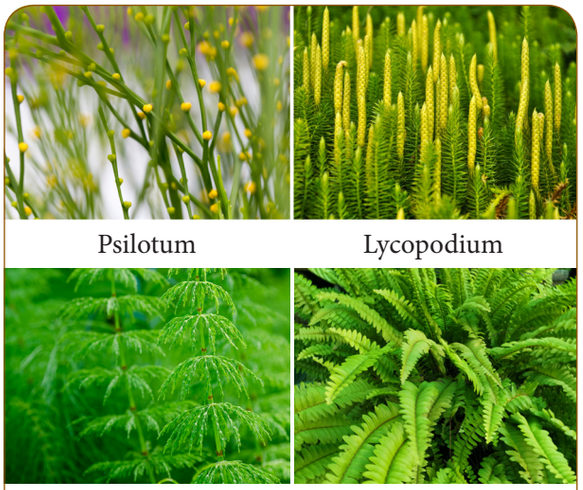
Pteridophytes also exhibit alternation of generation. The diploid sporophytic phase alternates with the haploid gametophytic phase. Sporophyte is the dominant phase. Sporophytes reproduce by means of spores. Spores are produced in sporangium. The sporangia bearing leaves are called sporophyll. Most of the plants produce only one type of spore either microspore or megaspore (homosporous). In some plants both microspore and megaspore are produced (heterosporous).

Spores give rise to gametophytic generation called *prothallus*, which is short lived and independent. The gametophytes produce the multicellular sex organs, antheridium which produces antherozoid (male gamete) and archegonium which contains an egg (female gamete). The antherozoid fertilizes with egg and form diploid zygote. It develops into an embryo which is differentiated into sporophyte.

17.4.1 Classification of Pteridophytes

Pteridophytes are classified into four classes. They are:

1. Psilopsida (Eg. *Psilotum*)
2. Lycopsida (Eg. *Lycopodium*)
3. Sphenopsida (Eg. *Equisetum*)
4. Pteropsida (Eg. *Nephrolepis*)



Psilotum | Lycopodium

Equisetum | Nephrolepis

Figure 17.10 Pteridophytes

More to Know

Lycopodium, is known as club moss.
Equisetum is known as horse tail.

17.4.2 Economic importance of Pteridophytes

- Ferns are used as ornamental plants.
- The rhizome and petioles of *Dryopteris* yield the vermifuge drug.
- The sporocarp of *Marsilea* (Water fern) is used as food by some people.

Table 17.5 Difference between Bryophytes and Pteridophytes

Bryophytes	Pteridophytes
Plant body cannot be differentiated into root, stem and leaf.	Plant body can be differentiated into root, stem and leaf.
Bryophytes are amphibians.	Pteridophytes are true land plants.
Vascular tissues are absent.	Vascular tissues are present.
The dominant phase of the plant body is gametophyte.	The dominant phase of the plant body is sporophyte.
Sporophytic generation depends on the gametophytic generation. E.g. *Riccia*	Gametophytic generation does not depend on sporophytic generation. Eg. *Selaginella*

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